

Primary Manufacturing Process

It involve the initial conversion of raw materials into useable forms. These processes typically shapes material without removing material & include

- Casting - shaping material by pouring
- Forming - changing the shape using mechanical force
- Joining - combining two metal pieces.

* Metal casting - one of oldest manufacturing process

steps - melting the metal.

→ Pouring the molten metal into the mold

→ Cooling & solidifying it in the shape of mold

→ Removing the final solidified product from the mold.

Tools Pattern - Replica of final casting product used to create mold cavity.

• molding box -

- Cope → Upper box
- Dreg → Bottom box
- Cheek → middle part.

→ container.

- molding sand - silica sand, water, clay - ...

- Riddle → Used to remove foreign particles.

- Rammer → used to compact sand.

- Sprue pin → Create sprue  for pouring molten metal

- Spoke pin →
- Vent rod → create holes for passing gas.
- Air → air and clean-lime sand.

- Bellow → blow air and clean loose sand.

- Crucible → melting the metal.



• Advantage - can produce complex shapes

- Suitable for large components
- Economical for mass production.

• Patterns - Single-piece pattern = simplest shapes

Split pattern = complex parts, made in two halves.

Match Plate pattern = pattern mounted on both side of plate

Cope & drag pattern = Cope & drag are separately

* molding - involves creating cavity that ~~take~~ hold molten metal. most common method → sand molding.

Type → Green sand, Dry sand, Shell mold,

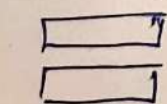
Furnace Cupola Furnace - mainly used for cast iron
 Induction Furnace - use electromagnetic induction
 Electric Arc Furnace - melt steel with electric arc
 * Crucible Furnace - non-ferrous metal.

* Metal Forming - :- metal forming changes the shape and size of a material through plastic deformation. - No material is removed. It often improves strength & grain structure of metal.

* Forging - It involves hammering or pressing the metal while it's hot or ~~not~~ cold.

Types - Open-die forging - hammering on flat surface
 Closed-die forging - Metal is pressed into dies shaped like the final product.

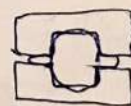
Application - crankshaft, gears, spanners, connecting rods.



flat dies.



(open)



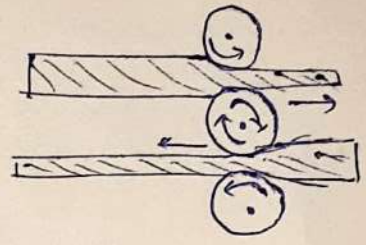
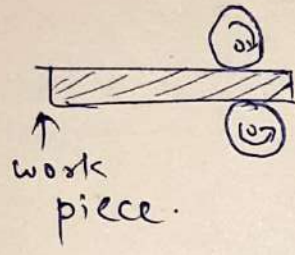
closed.

* Rolling - Process of reducing the thickness of changing the cross-section of a long workpiece by compressive forces applied through rollers.

Hot rolling - Above recrystallization temperature.
Easier shaping.

Cold rolling - Below recrystallization temperature.
Better surface, finish & strength.

Product - Sheets, beams, rails.



* Extrusion - Metal is forced through a die opening to form objects of a constant cross section.

└ Direct - Metal and ram move in same direction.

└ Indirect - Die moves towards stationary metal.

Product - Aluminum window frame, rods, tubing.

* Drawing - Drawing is used to pull metal through a die to reduce its diameter & increase length.

Types - wire drawing - making wire of copper, aluminum.

Tube drawing - Reducing diameter of pipes.

use - Electrical wire, bicycle spokes, seamless tube.

Tools → Hammer, anvil, Tongs, Swage block, forging dies, Rolling mill, Rollers,

Ram or plunger, container, Drawing die,

Drawing bench, Drum.

* Metal Joining Process *

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metal joining involves combining two or more metal parts to form a single unit. It may involve heating, pressure or filler material.

* welding :- Joining metals by heating to the melting point & fusing them together.

Types - Arc welding

SMAW (Shielded Metal Arc welding)

MIG (Metal inert gas)

TIG (Tungsten inert gas welding)



Gas welding

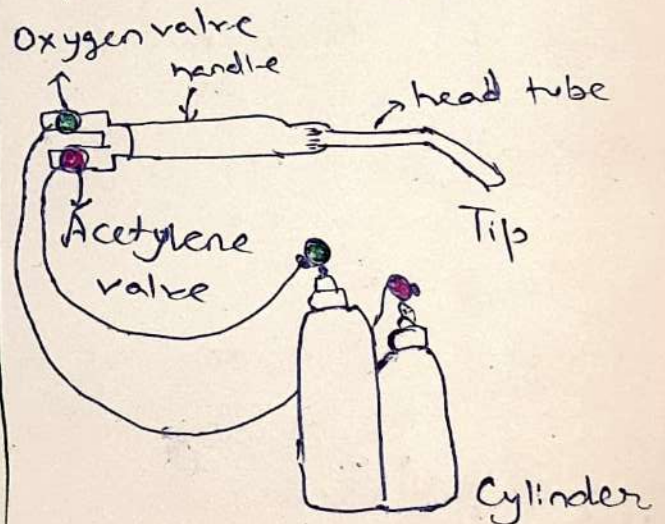
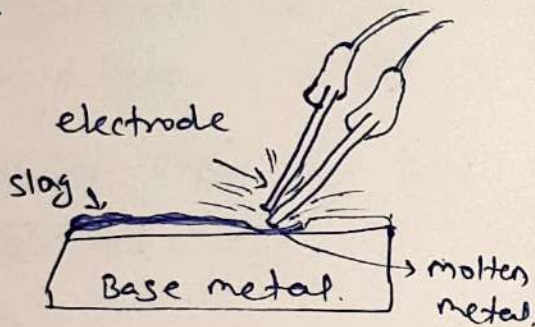
Oxy-acetylene flame used for cutting & joining.



In one cylinder O_2 gas is stored & in another, acetylene is stored. When both gases are mixed at high pressure and temp, it produce heat spark ($3200^\circ C$).

SMAW welding

mannual arc welding process that uses electrode coated with flux.



use to cut thick metal plates.

* Brazing Brazing joins metals using a filler metal with a melting point above 450°C but below the base metal. (capillary)
→ plumbing, bicycle frames.

* Soldering - similar to brazing but uses a filler metal with a melting point below 450°C
(Tin-lead alloy)
→ delicate jewelry, electrical connection.

Difference

<u>Brazing</u>	<u>Soldering</u>
1. melting temp. of filler above 450°C	melting temp. of filler below 450°C
2. <u>Filler metal</u> - copper alloy, silver alloy	<u>Filler metal</u> → lead tin
3. Fluxes → borax	Fluxes → zinc chlorides
4. Better strength of joint	less strength.